Gourav Kumar Ojha

AI/ML Engineer

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SUMMARY

AI/ML Engineer with hands-on experience building production-grade GenAI applications, cloud-native microservices, and distributed data pipelines. Proficient in Python (Pandas, NumPy, Scikit-learn, PyTorch), Go (working knowledge), TypeScript, LLM-driven workflows (LangChain, RAG, agentic systems), FastAPI, Docker, Kubernetes, and CI/CD on GCP, 3x Hackathon Winner, PUBLICATIONS (UNDER REVIEW) A Multi-Instrument Analysis Using Curiosity Rover and MCS Observations (MY32–MY36)in collaboration with IIRS(ISRO)

EDUCATION

Rungta College of Engineering and Technology — Bhilai, India
B.Tech in Computer Science (Artificial Intelligence), GPA: 7.8/10 — *Aug 2023 – Present*

TECHNICAL SKILLS

ML / AI: LLM Workflows, RAG, Agentic AI, Prompt Engineering, LangChain, PyTorch, TensorFlow, Hugging Face, Scikit-learn, NumPy, Pandas, OpenCV, SHAP, LIME, FAISS, Vector Databases, Jupyter Notebooks
Languages: Python, Go (working knowledge), TypeScript, JavaScript, C/C++, SQL
Cloud & DevOps: GCP (Vertex AI, BigQuery, Cloud Functions), Docker, Kubernetes, Microservices, Distributed Computing, Cloud-Native Applications, CI/CD, Release Management, Git, Prometheus
Backend & Data: REST API Development, FastAPI, Flask, React, Next.js, PostgreSQL, MongoDB
Engineering Practices: Security-First Mindset, Test Automation, Boundary Analysis, End-to-End Workflow Validation, Root Cause Analysis, Observability and Tracing, Design Thinking

WORK EXPERIENCE

Research Collaborator – Martian Dust Storm Analysis — Aug 2025 – May 2026
Indian Institute of Remote Sensing (ISRO), Dehradun — *Remote*

- Achieved **90–95%** dust storm detection accuracy by architecting a distributed, cloud-native Python data pipeline (Pandas, NumPy) processing **10TB+** across MRO/MCS, Curiosity (REMS/SAM), and Hubble datasets; validated with unit and integration tests using boundary analysis and equivalence class partitioning across 5 instrument sources.
- Reduced multi-instrument data reconciliation errors by debugging ML model outputs in Jupyter notebooks and implementing structured failure-recovery and root-cause-analysis workflows across 4 Martian years of atmospheric data.

Co-Founder & AI Engineer – AstraTech (Defence-Tech Startup, Team of 3) — Jan 2026 – Jul 2026
Rungta Business Incubation RIU, Bhilai — *Remote*

- Improved sensor-fault detection speed by **30–40%** by architecting a microservices-based, security-first embedded AI system with real-time sensor fusion (LWIR, mmWave, seismic) and a distributed CI/CD release-management pipeline with automated test stages and observability logging.
- Reduced deployment defects across releases by applying full-SDLC agile delivery and design-thinking-driven requirements analysis across hardware/software teams for role-based access and end-to-end workflow validation; design patent filed (Patent Pending).

KEY PROJECTS

SetuDrishti – AI-Powered Clinical Decision Support System — [Live]
PyTorch, HuggingFace, LangChain, FastAPI, PostgreSQL, Docker

- Achieved **98%** diagnostic accuracy by architecting a full-stack, cloud-native NLP and multi-modal deep learning platform with RAG-based EHR retrieval and role-based authentication, built as independently deployable microservices.
- Reached **85%** model accuracy across patient cohorts by building a distributed SHAP/LIME evaluation harness measuring output quality, defect detection rate, and consistency.

LUMEN AI – Enterprise Financial Assistant — [GitHub]
Multi-Agent RAG, GCP Vertex AI, BigQuery

- Cut test turnaround by **60%** by architecting a distributed, multi-agent RAG microservices pipeline on GCP managed services (Vertex AI, BigQuery) with version-controlled prompt deployment and full observability tracing.
- Reduced root-cause-analysis time by **10%** by building a security-first, cloud-native structured logging and tracing layer for transparent multi-agent execution.

HireMind AI – Enterprise Candidate Intelligence Platform — [Live]
Python, FastAPI, FAISS, Docker, Prometheus

- Delivered **34ms** startup and **120ms** retrieval latency at 12.5 RPS and 320MB peak memory by architecting a full-stack, containerized microservices platform using Hybrid Semantic Retrieval and a deterministic Explainability Engine.
- Cut deployment vulnerabilities by **30%** by implementing a security-first, multi-stage Docker release-management pipeline with Prometheus observability and distributed monitoring.

ACHIEVEMENTS

- **1st Place, E-Summit Startup Pitching Competition**, Vyom Cultural Fest, RISU (Mar 2026): Won top prize pitching AstraTech in a Shark Tank-style competition, awarded by Padma Shri Anand Kumar.
- **1st Place, Hack-Sprint Hackathon**, GDG on Campus RCET & Infinity Coders (Mar 2026): Led Team Alien X to build an AI-powered real-time Sepsis & Deterioration Monitoring System in a 12-hour sprint.

CERTIFICATIONS

- **Coding Spoon:** Generative AI, Agentic Systems & MLOps Deployment (Jun 2026) , **Google Cloud Skills Boost:** GCP & Applied ML (2025)
- **Microsoft Azure Fundamentals (AZ-900)**, **Postman API Expert**, **AICTE AI & ML Fundamentals**